```
clc;
clear;
close all;
```

PARAMETERS

```
A = 1;
f = 50;
Fs = 1000;
N = 1000;

n = 0:N-1;
x = A*sin(2*pi*f*n/Fs);

B = 8;
bits = [4 8 12 16];
```

8-BIT QUANTIZATION

```
D = 2*A/(2^B);

xr = D*round(x/D);
xt = D*fix(x/D);

er = x - xr;
et = x - xt;

SQr = 10*log10(mean(x.^2)/mean(er.^2));
SQt = 10*log10(mean(x.^2)/mean(et.^2));
```

INPUT AND OUTPUT QUANTIZATION

```
xq = xr;
y = x;

yq = D*round(y/D);
eo = y - yq;

SQout = 10*log10(mean(y.^2)/mean(eo.^2));
```

WORD-LENGTH COMPARISON

```
step = zeros(size(bits));
sqnrR = zeros(size(bits));
sqnrT = zeros(size(bits));
maxER = zeros(size(bits));
maxET = zeros(size(bits));

for k = 1:length(bits)
```

```
    b = bits(k);
    d = 2*A/(2^b);

    qr = d*round(x/d);
    qt = d*fix(x/d);

    er1 = x - qr;
    et1 = x - qt;

    step(k) = d;

    sqnrR(k) = 10*log10(mean(x.^2)/mean(er1.^2));
    sqnrT(k) = 10*log10(mean(x.^2)/mean(et1.^2));

    maxER(k) = max(abs(er1));
    maxET(k) = max(abs(et1));

end
```

DISPLAY RESULTS

```
fprintf('\n=======================================\n');
fprintf('       QUANTIZATION RESULTS\n');
fprintf('=======================================\n');

fprintf('Amplitude = %.1f\n',A);
fprintf('Signal Frequency = %d Hz\n',f);
fprintf('Sampling Frequency = %d Hz\n',Fs);
fprintf('Number of Samples = %d\n',N);
fprintf('Main Word Length = %d bits\n',B);
fprintf('8-bit Step Size = %.7f\n',D);

fprintf('Rounding SQNR = %.4f dB\n',SQr);
fprintf('Truncation SQNR = %.4f dB\n',SQt);
fprintf('Output SQNR = %.4f dB\n',SQout);

fprintf('Maximum Rounding Error = %.6f\n',max(abs(er)));
fprintf('Maximum Truncation Error = %.6f\n',max(abs(et)));
```

```
=======================================
       QUANTIZATION RESULTS
=======================================
Amplitude = 1.0
Signal Frequency = 50 Hz
Sampling Frequency = 1000 Hz
Number of Samples = 1000
Main Word Length = 8 bits
8-bit Step Size = 0.0078125
Rounding SQNR = 48.9339 dB
Truncation SQNR = 45.2931 dB
Output SQNR = 48.9339 dB
```

```
Maximum Rounding Error = 0.003483
Maximum Truncation Error = 0.005744
```

FIRST 10 SAMPLES TABLE

```
T1 = table(n(1:10)',x(1:10)',xr(1:10)', ...
    er(1:10)',xt(1:10)',et(1:10)', ...
    'VariableNames',{'n','Original','Rounded', ...
    'RoundError','Truncated','TruncError'});

fprintf('\nFIRST 10 SAMPLES\n');
disp(T1);
```

```
FIRST 10 SAMPLES
    n    Original    Rounded    RoundError    Truncated    TruncError
    _    ________    _______    __________    _________    __________

    0          0          0             0          0             0
    1    0.30902     0.3125     -0.003483    0.30469     0.0043295
    2    0.58779    0.58594     0.0018478    0.58594     0.0018478
    3    0.80902     0.8125     -0.003483    0.80469     0.0043295
    4    0.95106    0.95312    -0.0020685    0.94531      0.005744
    5          1          1             0          1             0
    6    0.95106    0.95312    -0.0020685    0.94531      0.005744
    7    0.80902     0.8125     -0.003483    0.80469     0.0043295
    8    0.58779    0.58594     0.0018478    0.58594     0.0018478
    9    0.30902     0.3125     -0.003483    0.30469     0.0043295
```

WORD-LENGTH COMPARISON TABLE

```
T2 = table(bits',step',sqnrR',sqnrT',maxER',maxET', ...
    'VariableNames',{'Bits','StepSize','SQNR_Round_dB', ...
    'SQNR_Trunc_dB','MaxError_Round','MaxError_Trunc'});

fprintf('\nWORD-LENGTH COMPARISON\n');
disp(T2);
```

```
WORD-LENGTH COMPARISON
    Bits     StepSize     SQNR_Round_dB    SQNR_Trunc_dB    MaxError_Round
MaxError_Trunc
    ____    __________    _____________    _____________    ______________
_____________

      4          0.125       23.667           20.871
0.059017          0.087785
      8      0.0078125       48.934           45.293
0.003483          0.005744
     12     0.00048828       78.811           65.891
0.00011536        0.00042324
     16     3.0518e-05       99.697           91.599
```

```
1.382e-05        2.6516e-05
```

OUTPUT PARAMETER TABLE

```
Parameter = {
    'Quantized signal'
    'Quantization error'
    'SQNR'
    'Word-length comparison'
};

Observation = {
    sprintf('8-bit rounding, Delta = %.7f',D)
    sprintf('Maximum rounding error = %.6f',max(abs(er)))
    sprintf('Rounding = %.4f dB; Truncation = %.4f dB',SQr,SQt)
    'SQNR generally increases with word length'
};

Unit = {
    'Amplitude'
    'Amplitude'
    'dB'
    'Bits and dB'
};

T3 = table(Parameter,Observation,Unit);

fprintf('\nOUTPUT PARAMETER TABLE\n');
disp(T3);
```

```
OUTPUT PARAMETER TABLE
            Parameter
Observation                                Unit
    ____________________________
_____________________________________________    ________________

    {'Quantized signal'      }    {'8-bit rounding, Delta =
0.0078125'             }    {'Amplitude'  }
    {'Quantization error'    }    {'Maximum rounding error =
0.003483'              }    {'Amplitude'  }
    {'SQNR'                  }    {'Rounding = 48.9339 dB; Truncation =
45.2931 dB'}    {'dB'         }
    {'Word-length comparison'}    {'SQNR generally increases with word
length'     }    {'Bits and dB'}
```

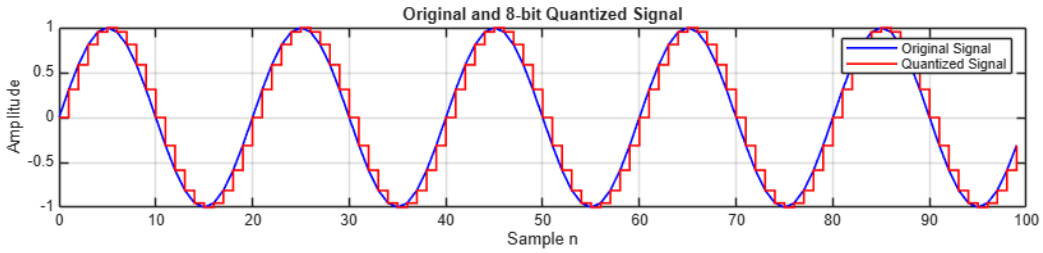
GRAPH 1: ORIGINAL AND QUANTIZED SIGNAL

```
figure;

plot(n(1:100),x(1:100),'b','LineWidth',1.2);
```

```
hold on;
stairs(n(1:100),xr(1:100),'r','LineWidth',1.2);

grid on;
xlabel('Sample n');
ylabel('Amplitude');
title('Original and 8-bit Quantized Signal');
legend('Original Signal','Quantized Signal');
```

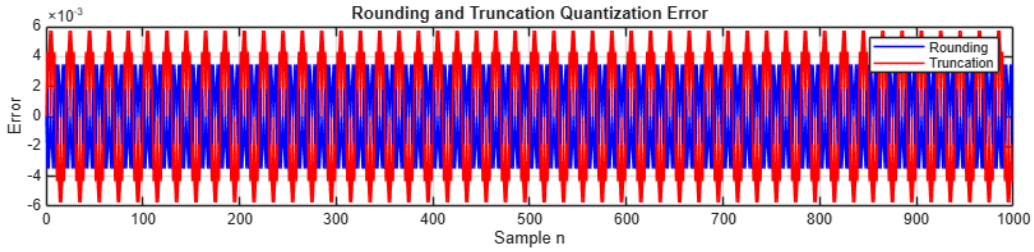


GRAPH 2: QUANTIZATION ERROR

```
figure;

plot(n,er,'b');
hold on;
plot(n,et,'r');

grid on;
xlabel('Sample n');
ylabel('Error');
title('Rounding and Truncation Quantization Error');
legend('Rounding','Truncation');
```



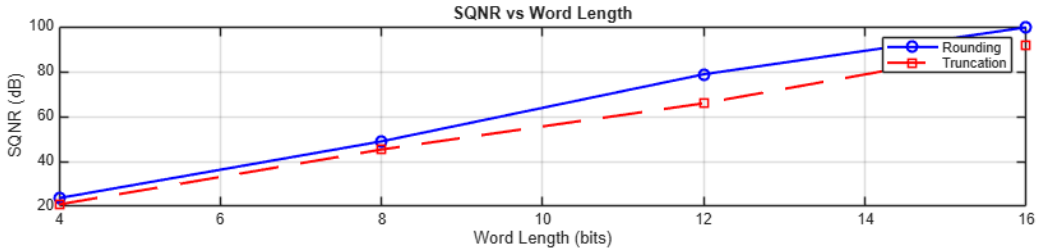
GRAPH 3: SQNR VS WORD LENGTH

```
figure;

plot(bits,sqnrR,'bo-','LineWidth',1.5);
hold on;
plot(bits,sqnrT,'rs--','LineWidth',1.5);

grid on;
xlabel('Word Length (bits)');
ylabel('SQNR (dB)');
```

```
title('SQNR vs Word Length');
legend('Rounding','Truncation');
```

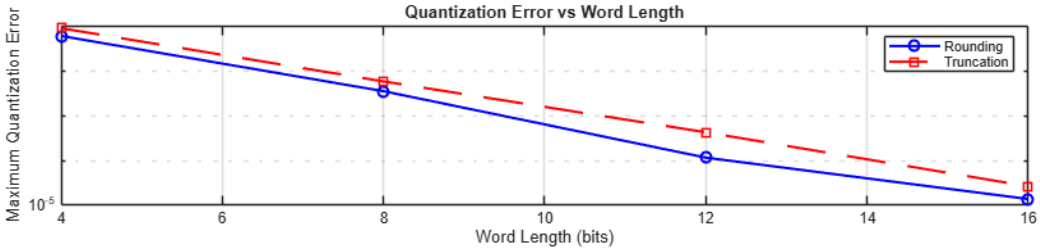


GRAPH 4: MAXIMUM ERROR VS WORD LENGTH

```
figure;

semilogy(bits,maxER,'bo-','LineWidth',1.5);
hold on;
semilogy(bits,maxET,'rs--','LineWidth',1.5);

grid on;
xlabel('Word Length (bits)');
ylabel('Maximum Quantization Error');
title('Quantization Error vs Word Length');
legend('Rounding','Truncation');
```



FINAL RESULT

```
fprintf('\n=======================================\n');
fprintf('             FINAL RESULT\n');
fprintf('=======================================\n');

fprintf('1. Uniform quantization was performed.\n');
fprintf('2. Rounding and truncation were compared.\n');
fprintf('3. Quantization errors were calculated.\n');
fprintf('4. SQNR was calculated for different word lengths.\n');
fprintf('5. Increasing word length generally improves SQNR.\n');
fprintf('6. All four graphs were generated successfully.\n');
```

```
=======================================
             FINAL RESULT
```

```
========================================
1. Uniform quantization was performed.
2. Rounding and truncation were compared.
3. Quantization errors were calculated.
4. SQNR was calculated for different word lengths.
5. Increasing word length generally improves SQNR.
6. All four graphs were generated successfully.
```

Published with MATLAB® R2026b